

# Contraction velocity influence the magnitude and etiology of neuromuscular fatigue during repeated maximal contractions

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This study aimed to compare the magnitude and etiology of neuromuscular fatigue during maximal repeated contractions performed in two contraction modes (concentric vs isometric) and at two contraction velocities (30/s vs 240°/s). Eleven lower limb-trained males performed 20 sets of maximal contractions at three different angular velocities: 0°/s (KE0), 30/s (KE30), and 240°/s (KE240). Cumulated work, number of contraction, duty cycle, and contraction time were controlled. Torque, superimposed and resting twitches, as well as gas exchange, were analyzed. Increasing contraction velocity was associated with greater maximal voluntary torque loss (KE0:  $-9.8 \pm 3.9\%$ ; KE30:  $-16.4 \pm 8.5\%$ ; KE240:  $-32.6 \pm 6.3\%$ ;  $P < 0.05$ ). Interestingly, the torque decrease was similar

for a given cumulated work. Compared with KE0, KE240 generated a greater evoked torque loss ( $\text{Db}_{100}$ :  $-24.3 \pm 5.3\%$  vs  $-5.9 \pm 6.9\%$ ;  $P < 0.001$ ), a higher  $\text{O}_2$  consumption ( $23.7 \pm 6.4 \text{ mL/min/kg}$  vs  $15.7 \pm 3.8 \text{ mL/min/kg}$ ;  $P < 0.001$ ), but a lower voluntary activation (VA) loss ( $-4.3 \pm 1.6\%$  vs  $-11.2 \pm 4.9\%$ ;  $P < 0.001$ ). The neuromuscular perturbations were intermediate for KE30 ( $\text{Db}_{100}$ :  $-10.0 \pm 6.8\%$ ; VA:  $-7.2 \pm 2.8\%$ ). Although the amount of mechanical work cumulated strongly determined the magnitude of torque decrease, the contraction velocity and mode influenced the origin of the neuromuscular fatigue. The metabolic stress and peripheral fatigue increased but reduction of VA is attenuated when the contraction velocity increased from 0°/s to 240°/s.

**Neuromuscular fatigue may be defined as the inability to maintain the required or expected torque output** (Taylor et al., 2005). The physiological mechanisms underpinning the development of neuromuscular fatigue have been investigated by manipulating exercise conditions such as duration and intensity of contraction(s), recovery duration, contraction velocity, and contraction mode (Boyas & Guevel, 2011). In particular, many researchers have examined the relationship between the muscle shortening velocity during maximal contractions and the magnitude of the decline in power after multi-joint exercise, i.e., cycling (Bogdanis et al., 2007; Tomas et al., 2010; Matsuura et al., 2011), or torque after mono-articular tasks (Mathiassen, 1989; Perry-Rana et al., 2002; Dalton et al., 2012). Most of these studies have observed greater fatigue at higher shortening velocities (Mathiassen, 1989; Perry-Rana et al., 2002; Bogdanis et al., 2007; Matsuura et al., 2011), although Dalton et al. (2012) reported a greater torque decrease for the lower velocity. These discrepancies could be due to the different protocols used since other parameters known to influence the fatigue magnitude also varied in these

studies, such as the number of contractions (Tomas et al., 2010) or contraction time and duty cycle (Monod & Scherrer, 1965; Mathiassen, 1989).

Neuromuscular fatigue may arise because of peripheral changes at the muscle level, i.e., peripheral fatigue, and/or because the central nervous system fails to drive the motoneurons adequately, i.e., central fatigue (Gandevia, 2001). Although contraction velocity seems to influence the etiology of fatigue (Perry-Rana et al., 2002; Babault et al., 2006; Matsuura et al., 2011), the manner in which this occurs is still disputed. For example, Matsuura et al. (2011) and Perry-Rana et al. (2002) compared cycling sprints performed at different contraction velocities and observed the greatest electromyographic (EMG) signal decrease, illustrating a motor drive perturbation, during the highest velocities. Conversely, Babault et al. (2006) observed greater EMG losses during isometric compared with concentric contractions, despite an identical level of torque loss. However, since the EMG signal can be altered by various factors, including a decrease in voluntary activation (VA) and/or sarcolemma excitability (Millet

et al., 2003), this technique cannot unambiguously establish the origin of fatigue. Other methods, such as the superimposed and potentiated resting twitches, are necessary to better identify the respective influence of peripheral and central fatigue (Millet et al., 2012).

Using these techniques, previous studies have focused on the influence of contraction mode on the origin of fatigue (Lanza et al., 2004; Babault et al., 2006; Callahan et al., 2009). To date, however, the results have not been consistent with Babault et al. (2006) and Lanza et al. (2004) who observed that concentric contractions induced greater alterations in the contractile properties compared with isometric contractions, while Callahan et al. (2009) did not observe any difference. Similarly, Babault et al. (2006) observed that central fatigue was predominant during isometric exercises, while Callahan et al. (2009) and Lanza et al. (2004) found no VA failure. Once again, differences in the protocols in terms of duty cycle or contraction time could be at the origin of these inconsistent results.

In addition to contraction mode, the etiology of fatigue may be influenced by the contraction velocity, which is known to affect muscle activation (Babault et al., 2002) and energy demands (Dorel et al., 2003). Indeed, Babault et al. (2002) suggested the presence of inhibitory mechanisms from joint receptors and Ib afferents, leading to a lower activation level, during single, maximal-intensity contractions performed at a low (60°/s) compared with a higher (120°/s) velocity. Furthermore, a greater metabolic demand has been observed during maximal exercise performed at a higher contraction velocity (Dorel et al., 2003). This may result in an accumulation of intracellular metabolites (ADP, Pi, and H<sup>+</sup>), which could impair the cyclical interaction of actin and myosin and Ca<sup>2+</sup> coupling processes (Allen et al., 2008; Debold, 2012). Thus, the influence of contraction velocity on muscle activation and metabolic stress could influence the etiology of fatigue.

Therefore, the purpose of the present study was to investigate the mechanisms of neuromuscular fatigue during maximal-intensity contractions performed in the isometric mode and at two isokinetic velocities (30°/s and 240°/s), with the same total contraction time and work-to-rest ratio. Neuromuscular (surface EMG, superimposed and potentiated resting twitches), as well as bioenergetic (gas exchange and blood lactate accumulation) techniques, were used to enhance the comprehension of the involved mechanisms. We hypothesized that high-velocity contractions would induce a higher energy demand (O<sub>2</sub> consumption and/or lactate accumulation) associated with more pronounced peripheral perturbations (i.e., larger evoked force decrease), while low-velocity and isometric contractions would induce a lower energy demand and muscle perturbation but higher central fatigue (i.e., larger VA decrease).

## Materials and methods

### Subjects

Eleven males volunteered to participate in this study [mean ± standard deviation (SD); age: 22.7 ± 1.9 year; mass: 87.6 ± 8.6 kg; height: 1.82 ± 0.06 m; training volume: 11.1 ± 3.1 h/week]. All were highly trained in the lower limbs and accustomed to performing repeated maximal efforts. Written informed consent was obtained from the subjects, and the study was conducted according to the declaration of Helsinki. Approval for the project was obtained from the local ethics committee of human research.

### Testing procedures

#### *Familiarization session*

The subjects first participated in a familiarization session, about 1 week prior to the first experimental session, to become accustomed with the testing procedures and to allow determination of the supramaximal electric-stimulation intensity to be used during all of the experimental sessions. During the familiarization session, subjects performed the warm up of the experimental sessions described below. Then, they were asked to perform an isometric maximal voluntary contraction (IMVC) of the knee extensors (KE) until they felt accustomed to the equipment. To determine each subject's supra-maximal stimulation intensity (I<sub>max</sub>), a series of single, square-wave stimuli were delivered (Digitimer DS7AH, Hertfordshire, UK) by progressively increasing the current until there was no further increase in the evoked isometric twitch response or the amplitude of the vastus lateralis (VL) M-wave. The last intensity obtained was increased by 50% (I<sub>max</sub>: 67 ± 18 mA). Thereafter, the subjects performed the complete series of neuromuscular tests (see below and Fig. 1b), a trial that included the three contraction velocities to be used in the experimental sessions (three sets of each), and another neuromuscular test procedure.

#### *Experimental sessions*

Three angular velocities (0°/s, 30°/s, and 240°/s) were tested on an isokinetic dynamometer (Biodex 3, Biodex, Shirley, New York, USA) during three sessions that were randomly assigned and separated by at least 48 h. The subjects performed 20 maximal knee extension sets, beginning every 30 s. The same 3 s of accumulated contraction time was performed within each set for each angular velocity: one contraction at 0°/s, knee angle: 90° (KE0; 0° is full extension); one contraction at 30°/s, range of motion: 105°–15° (KE30); eight contractions at 240°/s, range of motion: 105°–15° (KE240). For KE30 and KE240, to avoid any flexion work from the subjects, the experimenter manually generated the flexion movement at the same fixed velocity for the fatigue protocol. During this condition, the total work time represented 3 s; however, the time between the end of the eighth contraction and beginning of the next set represented 24 s added with the 3 s of total rest during the passive flexion movements (Fig. 1a). During all voluntary contractions, the subjects were asked to push as hard and fast as possible and were strongly encouraged. A visual torque feedback was also provided on a screen in front of the subject with a torque target corresponding to the maximal value measured in each situation. A typical experimental session was conducted as follows (see Fig. 1a): (a) preparation of the subject (electrode placement, mask and gas analysis system set up, verification of the signals); (b) warm up (10 min of cycling at 1 W/kg of body mass; isokinetic warm up: 8 × 240°/s; 6 × 180°/s; 4 × 90°/s; 2 × 30°/s; 2 × 0°/s); (c) 3 min of rest; (d) neuromuscular tests pre-fatigue in isometric mode (Fig. 1b); (e) 3 min of rest, including blood micro sampling for lactate analysis; (f) fatiguing procedure on the isokinetic dynamometer (20 sets); (g) neuromuscular tests

post-fatigue in isometric mode (starting exactly 24 s after the last contraction); and (h) blood micro sampling for lactate analysis (3 min after the end of the fatiguing procedure).

### Neuromuscular tests

The femoral nerve was stimulated percutaneously using a self-adhesive cathode (10-mm diameter, Ag-AgCl, Contrôle Graphique Medical, Brie-Comte-Robert, France) pressed manually over the femoral triangle. The anode, a 10 × 5 cm self-adhesive stimulation electrode (Medicompex SA, Ecublens, Switzerland), was located in the gluteal fold. Square-wave stimuli of 200  $\mu$ s duration with a maximal voltage of 400 V were delivered with a constant current.

Subjects were requested to perform three IMVC (duration: about 4 s, rest: 60 s) with the knee at 90°. During each IMVC, a high-frequency doublet stimulus (100 Hz) was delivered on the force plateau (visually controlled). The subject was asked to continue the IMVC after the stimulation for about 2 s and then to relax. A set of three stimulations were performed on the relaxed muscle in a potentiated state: one high-frequency doublet stimulation (100 Hz; 2 s after the IMVC), one low-frequency doublet stimulation (10 Hz; 5 s after the first stimulation), and a single twitch (5 s after the second stimulation; Fig. 1b). To ensure that the neuro-stimulation was supra-maximal, for each session, the baseline values were compared with those obtained during the familiarization session and only differences < 5% were accepted.

### Measurements and data analysis

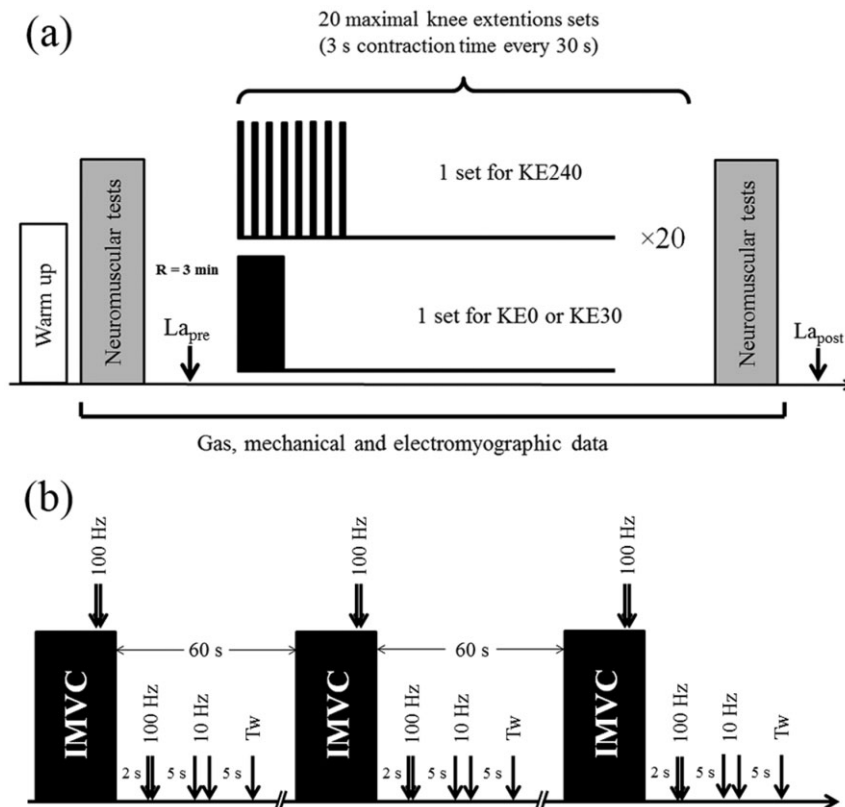
#### Torque

The torque was measured by the isokinetic dynamometer (sampling rate: 100 Hz) during the pre- and post-fatigue neuromuscular tests and the 20 all-out knee extension bouts. The subjects were seated with their hips at 80° and were tightly strapped across the chest, the hips, and the working leg to avoid lateral and frontal displacement. They were also asked to cross their arms over the chest during the voluntary contractions to limit the involvement of peripheral muscles.

During the neuromuscular tests, the IMVC plateau was determined over a 500-ms period prior to the superimposed twitch. The VA was calculated with the high frequency doublet (100 Hz) superimposed twitch from the highest pre-fatigue IMVC (i.e., best of the three sets) and the first post-fatigue IMVC to avoid any rest. The following equation was used (Eqn. [1]):

$$VA (\%) = [1 - (\text{superimposed twitch} / \text{potentiated twitch})] \times 100 \quad [1]$$

The maximal torque response on relaxed potentiated muscle for high-frequency (100 Hz) doublet stimulation ( $Db_{100}$ ), and low-frequency (10 Hz) doublet stimulation ( $Db_{10}$ ) were monitored. The ratio between  $Db_{10}$  and  $Db_{100}$  ( $Db_{10:100}$ ) was also calculated with the aim of establishing the possible predominance of low- or high-frequency fatigue (Millet et al., 2012). For  $Db_{100}$ ,  $Db_{10}$ , and  $Db_{10:100}$ , the average value from the three sets was considered.



**Fig. 1.** Experimental design (a) and neuromuscular testing procedure (b). The subjects performed one session for each of the three contraction velocities. The fatiguing protocol consisted in 20 sets of 3-s contraction time (i.e., one contraction for 0°/s – KE0 – and 30°/s – KE30 – and eight contractions for 240°/s – KE240) every 30 s.  $La_{pre}$  and  $La_{post}$  refer respectively to the lactatemia measurement pre- and post-fatigue. Arrows indicate the timing of motor nerve stimulation. Three type of stimulation were performed: doublet stimulation at 100 Hz, doublet stimulation at 10 Hz and single twitch (Tw). Stimulations were delivered during the isometric maximal voluntary contraction (IMVC) or on the relaxed potentiated muscle (baseline).

During the fatiguing protocol, the mean torque values of each set were considered and expressed as a percentage of the highest mean torque measured. For KE30 and KE240, the work was also calculated from the product of the mean torque value throughout one contraction and the angular amplitude of the movement (i.e., 90°). The torque decrease for a similar cumulated work and the number of contractions required to reach it was taken into consideration for each 1000 J step until 5000 J. The contraction considered was the one which led to the closest cumulated work from the considered step.

## Electromyography

During the neuromuscular tests, the VL EMG activity of the working leg was recorded using surface electrodes (EMG Triode, nickel-plated brass, electrode diameter = 1 cm, inter-electrode distance = 2 cm, Thought Technology, Montreal, Canada). EMG signals were collected using the Flexcomp Infiniti system (Thought Technology; 2048 Hz). The Flexcomp Infiniti system had an input impedance and common mode rejection ratio of 2 MΩ and > 110 dB, respectively. The EMG electrodes were positioned according to SENIAM recommendations (Hermens et al., 2000). The skin was shaved and cleaned with alcohol before placing electrodes to improve the contact between skin and electrode and to reduce skin impedance (< 2 kΩ). The electrodes were placed on the muscle belly and orientated longitudinally to the muscle fibers.

Raw EMGs were filtered (Butterworth order 2, bandpass 10–500 Hz) and amplified (gain = 500) before calculating root mean squared values (RMS) with a 50-ms moving rectangular window (Origin Pro 8.1, OriginLab, Northampton, Massachusetts, USA). The RMS<sub>max</sub> signals were the highest mean calculated over a 500-ms period before the stimulation during IMVC. The peak-to-peak amplitude of the VL M-wave responses to a single supra-maximal electrical stimulus were recorded (M<sub>max</sub>) and the average value from the three sets was considered. The RMS<sub>max</sub> was normalized to M<sub>max</sub> (RMS/M<sub>max</sub>). The highest RMS/M<sub>max</sub> during the three IMVCs was considered for each neuromuscular test and the percentage decrease was calculated between pre- and post-fatigue.

## Gas exchange

During the entire fatigue protocol, ventilation ( $\dot{V}_E$ ), oxygen consumption ( $\dot{V}O_2$ ), carbon dioxide output ( $\dot{V}CO_2$ ), and the respiratory exchange ratio (RER) were measured using a mobile gas analyzer (MetaMax 3b; Cortex Biophysik, Leipzig, Germany). The analyzer was warmed up and calibrated according to the manufacturer's instructions before each measurement. Gas analyses ( $\dot{V}_E$ ,  $\dot{V}O_2$ ,  $\dot{V}CO_2$ , and RER) were carried out breath by breath for the entire duration of each fatigue protocol. Afterwards, the values were averaged for each 30 s (i.e., one set of contractions and the following rest).

## Blood lactate

A micro blood sample was taken from the fingertip and analyzed for blood lactate concentration [La<sup>-</sup>] with the Lactate Pro (LT-1710, Arkray, Japan) portable analyzer just before the fatiguing procedure and after the neuromuscular test session post-fatigue (i.e., 3 min after the fatiguing protocol). The initial value was subtracted from the post-fatigue value to obtain the blood lactate accumulation (Maciejewski et al., 2012).

## Statistical analysis

All data were analyzed with Statistica 8.0 Software (Stat Soft Inc., Tulsa, Oklahoma, USA) and were expressed as means ± SD.

The normality of the error distribution was examined with Lilliefors' test. Homogeneity of variance was verified using Levene's test. With the assumption of normality and homogeneity of variance confirmed, a two-way analysis of variance (ANOVA) with repeated measures (pre- vs post-fatigue × velocity of contraction) was employed on each pre- vs post-variable, i.e., IMVC torque, VA, Db<sub>100</sub>, Db<sub>10:100</sub>, blood lactate accumulation, M<sub>max</sub>, and RMS<sub>max</sub>/M<sub>max</sub>. When a significant interaction was revealed, Tukey's post-hoc test was used to specify where the difference occurred, i.e., the pre-fatigue values were compared to ensure that they were not different, the post-fatigue values were compared with pre-fatigue to evidence the fatigue effect, and the post-fatigue values were compared between themselves to bring out the contraction velocity effect. A two-way ANOVA with repeated measures (cumulated work × velocity of contraction) was also employed on torque decrease and contraction number. Non-linear regression techniques were used to fit the kinetics of both torque (Hendrix et al., 2009) and gas variables ( $\dot{V}_E$ ,  $\dot{V}O_2$ , and  $\dot{V}CO_2$ ; Barstow & Molé, 1991) during the 20 sets with a mono-exponential function (Eqns. [2 and 3]).

$$\text{Torque}(t) = A + (100 - A) \times (1 - e^{(-t/\tau)}) \quad [2]$$

Torque and A (asymptote) are expressed as a percentage of the best mean torque measuring during the fatiguing protocol; τ, curvature constant (unit: set); t, time (unit: set)

$$\text{Gas variable}(t) = R + A(1 - e^{(-t/\tau)}) \quad [3]$$

R is the resting value and A is the asymptote, they are expressed in mL/min/kg for  $\dot{V}O_2$  and  $\dot{V}CO_2$ , and in L/min for  $\dot{V}_E$ ; τ, curvature constant (unit: set); t, time (unit: set)

An iterative process was used in order to minimize the sum of squared error between the fitted function and the observed values. A one-way ANOVA with repeated measures (contraction velocity) was employed to compare the different equation component (asymptote and curvature constant). The alpha level for statistical significance was set at  $P < 0.05$ .

## Results

### Neuromuscular properties

None of the pre-fatigue values were different between the three experimental sessions. The calculated cumulated work steps were not different for KE30 and KE240. IMVC decreased in all three conditions (Table 1 and Fig. 2a), and post-fatigue IMVC was significantly lower when the contraction velocity was higher. Torque loss during the IMVC was the lowest for KE0 and the highest for KE240 (Fig. 3a). The torque asymptotes calculated during fatiguing contractions were lower when the contraction velocity increased (KE0:  $88 \pm 4\%$ ; KE30:  $81 \pm 6\%$ ; KE240:  $66 \pm 7\%$ ;  $P < 0.05$ ; Figs 4 and 5). For the same cumulated work, until 5000 J, the torque decrease was similar between KE30 and KE240. Conversely, the number of contractions necessary to reach a similar cumulated work was greater for KE240 compared with KE30 ( $P < 0.05$ ; Fig. 6). The Db<sub>100</sub> decrease was greater after KE240 ( $-24.3 \pm 5.3\%$ ) was compared with KE30 ( $-10.0 \pm 6.8\%$ ;  $P < 0.01$ ). During KE0, Db<sub>100</sub> did not significantly change ( $-5.2 \pm 6.9\%$ ;  $P = 0.37$ ; Figs. 2b and 3b). VA was significantly lower after the fatiguing protocol for KE0 ( $-11.2 \pm 4.9\%$ ;  $P < 0.001$ )



Table 1. Neuromuscular properties before and after fatigue for each contraction velocity

|                         | KE0           |                             | KE30          |                             | KE240         |                             |
|-------------------------|---------------|-----------------------------|---------------|-----------------------------|---------------|-----------------------------|
|                         | Before        | After                       | Before        | After                       | Before        | After                       |
| IMVC (N·m)              | 322 ± 86      | 291 ± 71* <sup>‡§</sup>     | 322 ± 91      | 269 ± 93* <sup>‡§</sup>     | 320 ± 101     | 215 ± 62* <sup>‡†</sup>     |
| VA (%)                  | 96.4 ± 3.1    | 85.4 ± 6.9* <sup>‡§</sup>   | 95.6 ± 3.1    | 88.8 ± 6.4* <sup>‡§</sup>   | 96.4 ± 4.0    | 92.3 ± 4.9 <sup>†‡</sup>    |
| Db <sub>100</sub> (N·m) | 117 ± 21      | 111 ± 30 <sup>c</sup>       | 115 ± 17      | 103 ± 21* <sup>§</sup>      | 121 ± 21      | 92 ± 13* <sup>†‡</sup>      |
| Db <sub>10:100</sub>    | 0.88 ± 0.06   | 0.83 ± 0.08                 | 0.89 ± 0.16   | 0.84 ± 0.10                 | 0.89 ± 0.11   | 0.64 ± 0.08* <sup>†‡</sup>  |
| M <sub>max</sub> (mV)   | 6.12 ± 2.48   | 6.02 ± 2.42                 | 6.93 ± 4.92   | 6.85 ± 3.87                 | 6.34 ± 2.87   | 6.13 ± 3.36                 |
| RMS/M <sub>max</sub>    | 0.115 ± 0.045 | 0.095 ± 0.038* <sup>§</sup> | 0.117 ± 0.061 | 0.097 ± 0.044* <sup>§</sup> | 0.106 ± 0.035 | 0.111 ± 0.029 <sup>†‡</sup> |

Values are mean ± standard deviation.

\*Significant difference with pre-fatigue.

<sup>†</sup>Significant difference with KE0.

<sup>‡</sup>Significant difference with KE30.

<sup>§</sup>Significant difference with KE240.

Db<sub>10:100</sub>, ratio between peak torque at 10 Hz and 100 Hz doublet stimulation; Db<sub>100</sub>, peak torque during 100 Hz doublet stimulation; IMVC, isometric maximal voluntary torque; M<sub>max</sub>, M-wave peak-to-peak maximal amplitude; RMS/M-wave, RMS value normalized by M<sub>max</sub>; VA, voluntary activation.

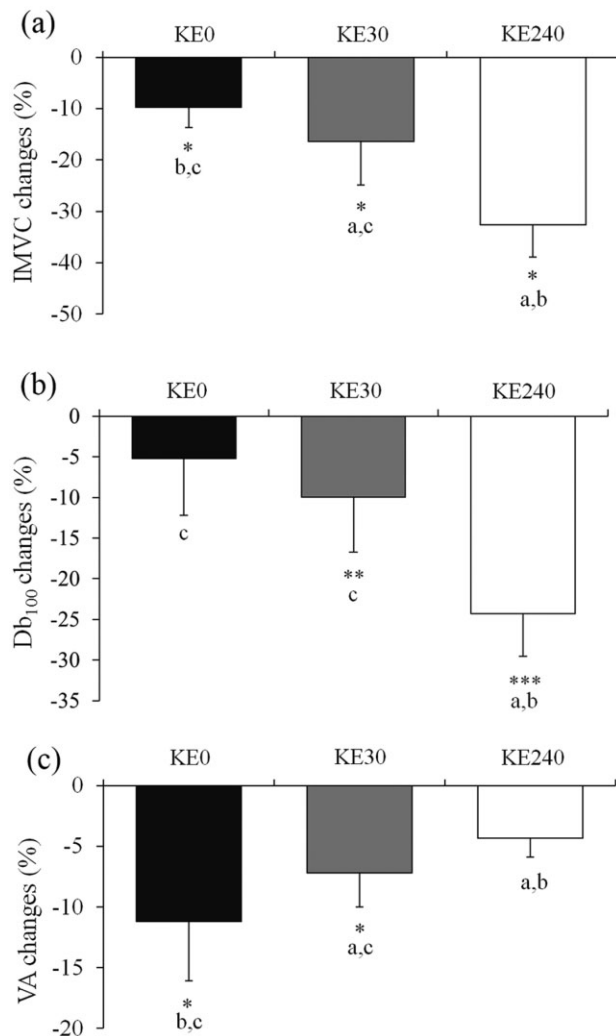


Fig. 2. Changes in maximal isometric voluntary torque (IMVC, a), evoked torque (Db<sub>100</sub>, b) and voluntary activation (VA, c) between pre- and post-fatigue tests; significant changes pre- vs post-fatigue: \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ ; (a) significantly different from KE0; (b) significantly different from KE30; (c) significantly different from KE240.

and for KE30 ( $-7.2 \pm 2.8\%$ ;  $P < 0.001$ ), but not for KE240 ( $-4.3 \pm 1.6\%$ ;  $P = 0.08$ ). Furthermore, the VA decrease was greater for KE0 compared with KE30 ( $P < 0.01$ ; Figs. 2c and 3a). Db<sub>10:100</sub> shifted downward only for KE240 ( $P < 0.001$ ), while M<sub>max</sub> did not decrease in any condition (Table 1 and Fig. 3c). RMS/M<sub>max</sub> decreased for KE0 and KE30 ( $P < 0.05$ ), but not for KE240 ( $P = 0.78$ ; Table 1).

#### Metabolic data

The lactate accumulation was higher for KE240 compared with KE0 and KE30 ( $P < 0.01$ ), but was not different between KE0 and KE30 ( $P = 0.09$ ; Table 2). The plateau of O<sub>2</sub> consumption and CO<sub>2</sub> output was similar between KE0 and KE30, and both were significantly lower than KE240 ( $P < 0.01$ ; Table 2). Moreover,  $\dot{V}_E$  was different among the three conditions ( $P < 0.05$ ; Table 2). The RER was also different between KE240 and KE30 ( $P < 0.05$ ), but KE0 was not different from both situations (Table 2).

#### Discussion

The present study has shown that, during repeated, maximal, isokinetic efforts, the cumulative work strongly determined the magnitude of fatigue. Conversely, differences in the origin of the neuromuscular fatigue (i.e., VA and/or contractile properties) were observed after identical contraction times and duty cycles between the contraction modes and velocities tested in this study. The highest contraction velocity allowed a greater work production and induced a higher metabolic demand (O<sub>2</sub> consumption, blood lactate accumulation), a greater torque decrease, and a more peripherally generated neuromuscular fatigue (larger Db<sub>100</sub> decrease). Conversely, the lower velocity and isometric contractions were associated with high force production

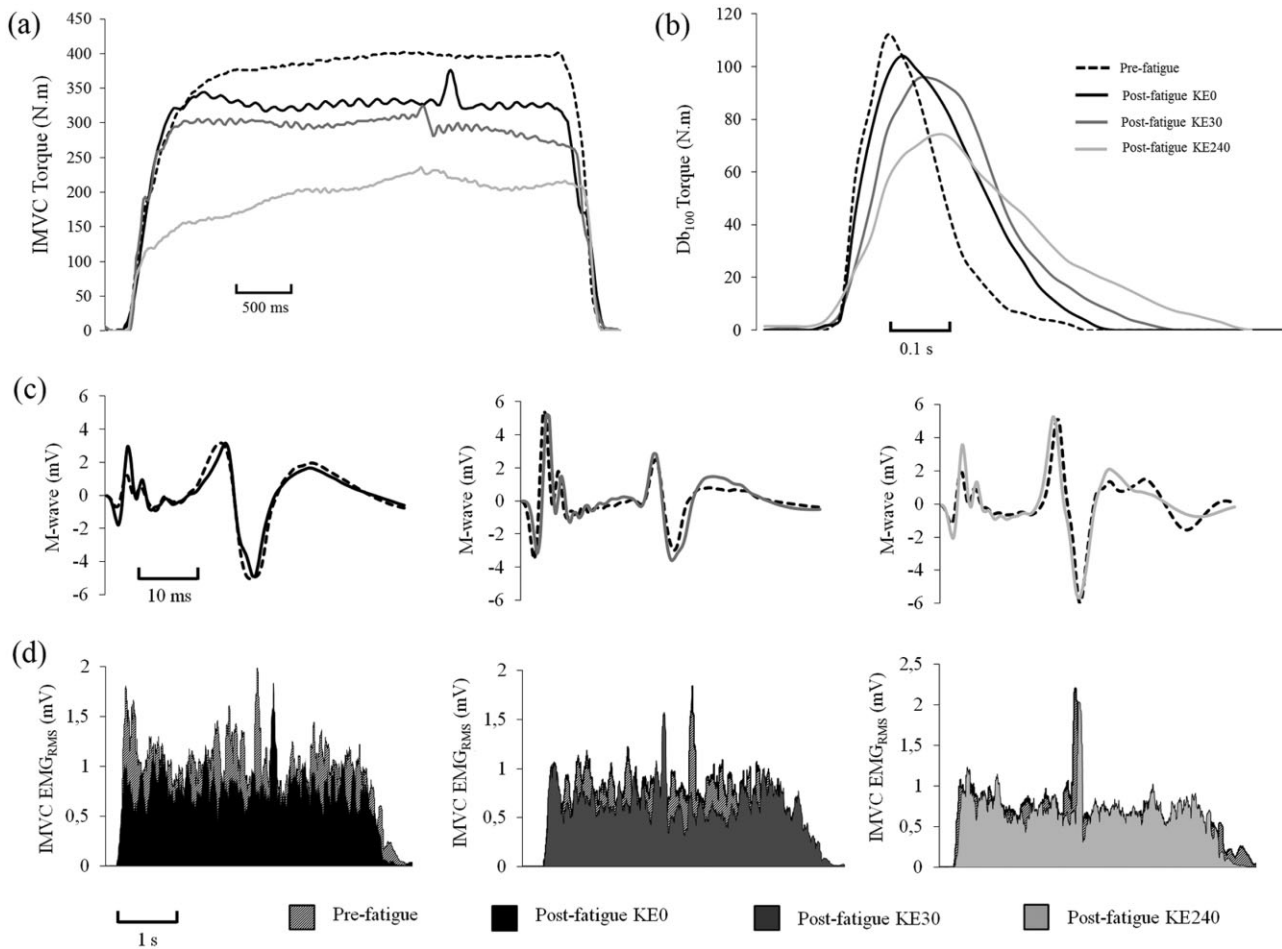


Fig. 3. Typical raw traces obtained during the neuromuscular testing: torque (a) and electromyographic response ( $EMG_{rms}$ ) of the VL during the IMVC (d), Db100 torque (b) and  $M_{max}$  (c) evoked on relaxed muscle, before (dotted line and hatched area) and after (KE0: black line and area; KE30 dark gray line and area; KE240 light gray line and area) the fatiguing protocol.

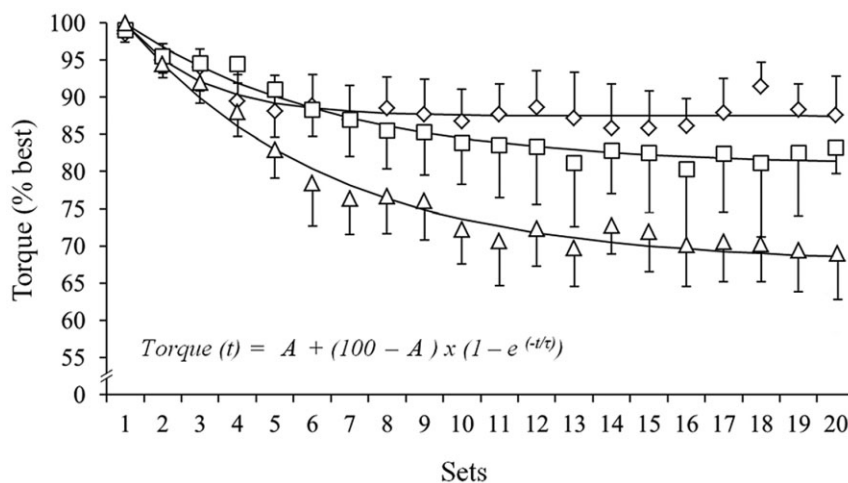


Fig. 4. Torque loss kinetics at the three contraction velocities during the fatiguing protocol. Data are expressed in percentage of the maximal torque measured within the fatiguing protocol for each subject. Dots represent the mean values with the associated SD; ◇, KE0; □, KE30; △, KE240.

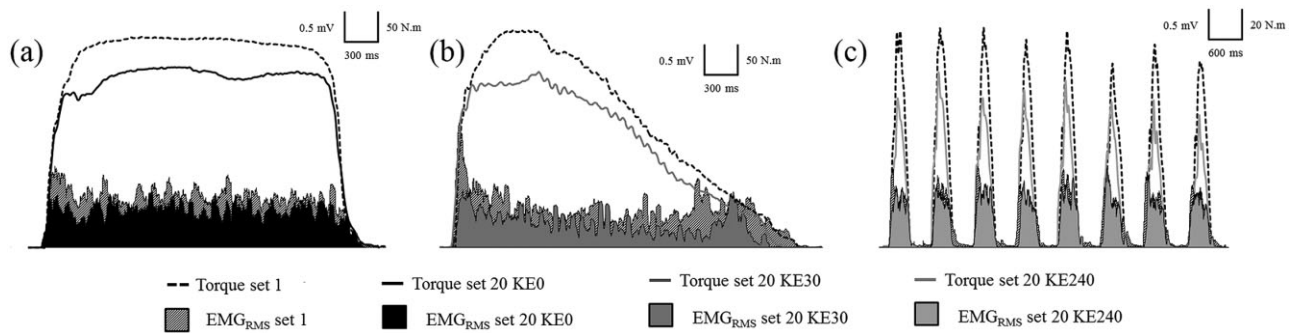


Fig. 5. Typical raw traces of torque and electromyographic response ( $EMG_{rms}$ ) of the VL obtained during the first (dotted line and hatched area) and the last set for KE0 (a) (black line and area), KE30 (b) (dark gray line and area), and KE240 (c) (light gray line and area).

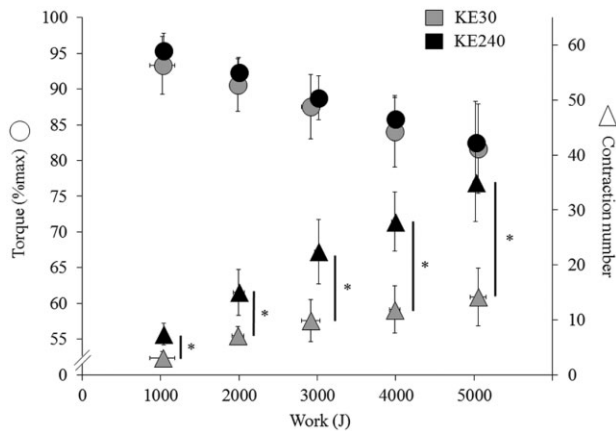


Fig. 6. Torque decrease (circle) and contraction number (triangle) depending on cumulated work. Data are mean values with associated SD for KE30 (gray) and KE240 (black). \*Significantly different  $P < 0.05$ .

Table 2. Metabolic data for each contraction velocity

|                           | KE0                        | KE30                       | KE240                      |
|---------------------------|----------------------------|----------------------------|----------------------------|
| $\dot{V}O_2$ (mL/min/kg)  | $15.7 \pm 3.8^\dagger$     | $15.7 \pm 4.9^\dagger$     | $23.7 \pm 6.4^{*\dagger}$  |
| $\dot{V}CO_2$ (mL/min/kg) | $17.7 \pm 5.6^\dagger$     | $15.0 \pm 5.4^\dagger$     | $26.3 \pm 7.0^{*\dagger}$  |
| $\dot{V}_E$ (L/min)       | $51.5 \pm 11.8^{*\dagger}$ | $37.7 \pm 11.1^{*\dagger}$ | $65.0 \pm 20.5^{*\dagger}$ |
| RER                       | $1.08 \pm 0.15$            | $0.94 \pm 0.12^\dagger$    | $1.13 \pm 0.09^\dagger$    |
| $\Delta[La^-]$ (mmol/L)   | $1.6 \pm 3.0^\dagger$      | $-1.0 \pm 0.9^\dagger$     | $5.7 \pm 4.0^{*\dagger}$   |

Data are mean plateaus (i.e., asymptote)  $\pm$  standard deviation for each gas variables.

\*Significant difference from KE0.

$^\dagger$ Significant difference from KE30.

$^\ddagger$ Significant difference from KE240.

$\Delta[La^-]$  is the blood lactate accumulation (lactatemia post-fatigue – pre-fatigue); RER, respiratory exchange ratio;  $\dot{V}CO_2$ , carbon dioxide output;  $\dot{V}_E$ , volume of ventilation;  $\dot{V}O_2$ , oxygen consumption.

and induced larger changes in VA (i.e., more central fatigue). Thus, the contribution of peripheral and central parameters was affected by the torque-velocity condition of the repeated contractions.

### Magnitude of fatigue

The torque losses observed after 20 bouts were greater when the contraction velocity was higher. This is in line with previous results obtained for cycling sprints (Tomas et al., 2010) and isokinetic contractions (Callahan et al., 2009), but is in opposition to the results of (Dalton et al., 2012). These discrepancies are not surprising since fatigue development is not only influenced by the contraction velocity, but also by numerous factors such as duty cycle, contraction time, contraction number, or work performed (Monod & Scherrer, 1965). In the study of Dalton et al. (2012), the resting time between two contractions was identical (0.6 s) for the low- and moderate-velocity condition (respectively 60°/s and 180°/s), but the contraction time, and thus the duty cycle, was three times higher for the low-velocity condition (1.5 s vs 0.5 s). This has probably contributed to the greater torque loss observed in this situation (Mathiassen, 1989).

In the same way, although the present study used an identical contraction time and duty cycle for all of the conditions tested, the number of contractions performed during the high-velocity condition (KE240) was eight times greater than both other conditions (KE0 and KE30). Thus, the number of contractions may have contributed to the higher fatigue amplitude for KE240 compared with KE30 and KE0 (Tomas et al., 2010). However, it is worth noting that the torque loss in KE30 remained higher than that observed at KE0 for an identical contraction time, duty cycle, and contraction number. Furthermore, Fig. 6 shows that for a similar accumulated work, KE30 and KE240 torque losses were identical despite participants performing twice the number of contractions during KE240. In conclusion, the present results showed that the mechanical conditions of maximal intermittent exercise (force-velocity characteristic and number of contractions) determine the external work that can be produced which, in turn, greatly influences the torque loss magnitude. The original contribution of the present study is also related to the investigation methods (EMG, electrical stimulation,  $O_2$

consumption, and lactate accumulation) that allowed the identification of the fatigue mechanisms linked to different contraction modes and velocities.

## Fatigue etiology – peripheral components

Considering that no significant differences were observed between the relative decrease in the torque produced during the intermittent efforts (between the first and last set), and the pre- vs post-fatigue IMVC changes, the neuromuscular tests seem to be representative of the fatigue phenomenon experienced in the three exercise conditions. The present data provide evidence of peripheral fatigue, indicated by the attenuation of the evoked torque ( $Db_{100}$ ), only for KE30 and KE240. This is consistent with the earlier and greater decrease in peak doublet amplitude previously reported during the repetition of concentric ( $60^\circ/s$ ) compared with isometric contractions (Babault et al., 2006). The greater evoked torque decrease for KE240 compared with KE30 could be related to the greater number of contractions since this is known to influence the torque decrease (Tomas et al., 2010). However, as discussed above, the cumulative work seemed to have a prevailing influence on the fatigue experienced. As a tentative explanation, despite the influence of contraction velocity on the mechanical efficiency, defined as the external work divided by the estimated energy consumption (Lodder et al., 1991), the energy consumption can be considered to be related to the mechanical work performed (Fenn, 1923; Wilkie, 1968). Thus, one may hypothesize that the energy demand was higher and induced larger local perturbations for KE240. This is consistent with the values for lactate accumulation and oxygen consumption, which were greater for KE240. Additionally, it is interesting to note that the oxygen consumption measured for the same cumulative work (5000 J) was higher during KE240 ( $20.8 \pm 4.9$  mL/min/kg) compared with KE30 ( $14.6 \pm 4.8$  mL/min/kg;  $P < 0.001$ ). This suggests that for a similar cumulative work, KE240 may have induced higher biochemical perturbations (i.e., increased  $H^+$ ,  $P_i$ , ADP) leading to greater peripheral fatigue (Allen et al., 2008; Debold, 2012). The 27% decrease in  $Db_{10:100}$  ratio provides evidence of low-frequency fatigue, which is generally attributed to alterations in  $Ca^{2+}$  excitation–contraction coupling processes (Millet et al., 2012). In contrast,  $M_{max}$  stayed unchanged which suggests that sarcolemma conduction capacity remained unaltered by the fatiguing protocols.

The evoked torque ( $Db_{100}$ ) decreased significantly for KE30, but did not change in the KE0 condition. Although no statistical difference was observed between the decrease in  $Db_{100}$  for KE30 and KE0, there was a trend for the KE30 decrease to be slightly higher than the KE0 decrease ( $P = 0.10$ ). This cannot be explained by alterations in sarcolemmal conduction capacity since  $M_{max}$  did not change after either KE0 or KE30. Thus, the

difference between KE0 and KE30 may be linked to perturbations in the cyclical interaction of actin and myosin and/or the  $Ca^{2+}$  coupling process. Indeed, despite the fact that whole-body  $O_2$  consumption was not significantly different between KE0 and KE30, it can be hypothesized that the energy consumed by the KEs was slightly lower for the isometric condition. Indeed, respiratory muscle work was very likely higher during KE0 than KE30 since  $\dot{V}_E$  was 37% greater (Coast et al., 1993). Similarly, the contribution of isometric contractions of the trunk muscles was probably higher during KE0 than KE30 since it increases with the torque level produced (Andersen & Saltin, 1985). In conclusion, it appears that peripheral components of fatigue are higher during the concentric mode when contraction velocity is increased. This seems to be partly explained by the higher cumulative work performed, which induces a higher energy consumption.

## Fatigue etiology – central components

Reduced EMG activity can occur from a reduced neural drive to the motoneurons, reflecting a central nervous system role in fatigue development (Gandevia, 2001) and/or from changes in sarcolemma excitability, i.e., peripheral fatigue (Pasquet et al., 2000). To evaluate the relative influence of these two parameters, we normalized RMS during an IMVC to  $M_{max}$  before and after the fatiguing protocol. This parameter showed a significant decrease for KE0 and KE30 only (Table 1), suggesting that isometric and low-velocity contractions lead to central motor drive perturbations. Similarly, significant decreases in VA were found for KE0 and KE30 only. Interestingly, VA changes were almost two times greater for KE0 compared with KE30 (Table 1). Different VA decreases have been previously reported for different contraction modes. Indeed, Babault et al. (2006) observed that fatigue was first central and then peripheral for isometric contractions, while the pattern was reversed for concentric contractions. Our study provides new data indicating that the VA decrease is likely to be more related to the contraction velocity than the contraction mode.

The present data cannot identify the location(s) of the impairment within the central nervous system, but a number of spinal and supra-spinal mechanisms could have played a role (for review, see Gandevia, 2001). Recent studies have focused on the influence of III/IV metabo-sensitive muscle afferents group and their effect on spinal (Amann et al., 2013) and supra-spinal (Taylor et al., 2000) inhibition. In the present study, VA during KE0 decreased nearly two times more than during KE30, whereas whole-body  $O_2$  consumption was similar. Isometric contractions might have limited the blood flow and create higher local disturbances, as shown by the high RER ( $1.08 \pm 0.15$ ), which could have stimulated the metabo-sensitive afferents. Moreover, high mechanical stress is also known to induce spinal inhibition



especially via Ib muscle afferents (Gandevia, 2001). Torque level during KE30 was two times lower than KE0. The stimulation of metabo- and mechano-sensitive afferents could therefore explain the VA inhibition for KE0 and KE30, and the difference between both.

The entire high-velocity contraction protocol did alter neither VA nor the sarcolemmal excitability after 20 bouts of maximal exercise. One can hypothesize that the underlying causes of fatigue were similar after only 20 contractions at KE240 or for a cumulative work production of 5000 J. The present results indicate that for a given number of contractions or cumulated work production, the fatigue origin was mainly peripheral for KE240 whereas it was combined (i.e., peripheral and central) for KE30. Therefore, the total work performed seemed to be the main determinant of the fatigue amplitude observed in the KE30 and KE240 protocols, but the force-velocity characteristic of the contraction generated different fatigue etiology in these two conditions. Interestingly, the isometric condition induced no significant modifications in the evoked torque ( $Db_{100}$ ), whereas VA was greatly depressed in this condition. This may be due to the maximal torque level produced in this condition which may have induced VA inhibition by the mechano-sensitive afferents.

The present study showed that, although the cumulative work performed seemed to largely explain the magnitude of torque decrease, the contraction velocity and mode also influenced the origin of the neuromuscular fatigue when contraction time and duty cycle were controlled. High-velocity contractions induced higher energy consumption and large changes in contractile properties, whereas low-velocity and isometric contractions induced higher VA inhibition. This indicates that

fatigue mechanisms are influenced by the force-velocity condition of the repeated contractions.

## Perspectives

It is important to note that the level of cumulated work and the number of contractions were manipulated separately, making it difficult to precisely evaluate the role played by each variable on the magnitude of the neuromuscular fatigue observed in this study. The central origin of fatigue during maximal intermittent contractions also remains to be specified in terms of localization, i.e., spinal vs supraspinal, and type of afferents implied, i.e., mechano- vs metabo-sensitive afferents. Furthermore, although the isokinetic device allowed us to control the contraction velocity and to focus on the influence of neuromuscular fatigue on torque production, it should be kept in mind that under ecological sport or daily life situations the movement velocity also tends to decrease with fatigue. The complex interactions and mechanisms of concomitant torque and velocity decrease leading to power loss remain a relevant topic and should be addressed, for example, with an iso-inertial device. Nevertheless, the present study offered an original approach of how the mechanical conditions influence the neuromuscular fatigue, and reinforces the importance of considering the contraction velocity for further studies related to fatigue magnitude and etiology.

**Key words:** Central fatigue, peripheral fatigue, intermittent exercise, isokinetic contraction, cumulated work.

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